Configuring the NX-400 Edge Router for Fixed-Wireless Sites

Document	KVT-DG-112
Revision	C, 12 October 2026
Applies to	NX-400 hardware revision 2 and later, firmware 7.4.x
Owner	Access Network Engineering

1. Purpose and Scope

This guide describes how to install and configure an NX-400 edge router at a fixed-wireless base station. It covers the initial configuration, the VLAN plan, monitoring and the checks to complete before a site is handed over to operations. Changes to the core network are outside its scope and follow the change process in KVT-OPS-004.

2. Before You Start

Bring the following to site:

- A laptop with an SSH client and the site configuration file from the provisioning portal.
- The site’s management IP address, gateway and VLAN identifiers.
- A console cable (RJ45 to USB-C) and a spare SFP+ module.
- A change ticket in the Approved state.

2.1 Safety

Never climb the tower or open the cabinet without a second person present. Report any damaged earth bonding before you begin work.

> **Warning.** Do not connect the radio uplink until the router reports a valid configuration. An unconfigured router floods the backhaul with DHCP requests.

3. Port Assignment

Port	Speed	Connected to	VLANs
ge-0/0/0	1 Gbit/s	Sector radio A (north)	210, 999
ge-0/0/1	1 Gbit/s	Sector radio B (south-east)	211, 999
ge-0/0/2	1 Gbit/s	Sector radio C (south-west)	212, 999
xe-0/1/0	10 Gbit/s	Microwave backhaul	100, 300
ge-0/0/7	1 Gbit/s	Site management switch	999

4. Initial Configuration

4.1 Load the base configuration

Connect to the console port at 115200 baud and log in with the factory account. Copy the site file to the router, check it with a dry run, then apply it:

```
scp KV-0417-base.yaml admin@192.0.2.1:/config/
ssh admin@192.0.2.1
nxcli config load /config/KV-0417-base.yaml --dry-run
nxcli config load /config/KV-0417-base.yaml --commit --comment "KVT-CHG-30218"
```

The dry run must finish without errors before the commit is attempted.

4.2 Site configuration file

The site file is generated by the provisioning portal. A shortened example is shown below.

```
site:
  id: KV-0417
  name: "Harrow Ridge"
  timezone: Europe/London
management:
  vlan: 999
  address: 10.41.7.2/29
  gateway: 10.41.7.1
uplink:
  interface: xe-0/1/0
  vlans: [100, 300]
  mtu: 9100
sectors:
  - { port: ge-0/0/0, vlan: 210, azimuth: 0 }
  - { port: ge-0/0/1, vlan: 211, azimuth: 120 }
  - { port: ge-0/0/2, vlan: 212, azimuth: 240 }
```

5. VLAN Plan

VLAN	Name	Purpose	Notes
100	BH-DATA	Subscriber traffic to the aggregation ring	Tagged on the backhaul only
210–212	SECT-A to SECT-C	Subscriber traffic per sector	One VLAN per sector radio Never bridge two sectors
300	BH-OAM	Backhaul radio management	Reachable from the NOC only
999	SITE-MGMT	Router, switch and radio management	Untagged on ge-0/0/7

6. Monitoring

The router exports its metrics to the regional collector every 60 seconds. The telemetry section of the configuration must read as follows:

```
[telemetry]
collector = 10.0.88.20:4317
interval_s = 60
export = interfaces, optics, cpu, temperature

[alarms]
optics_rx_low_dbm = -18.0
temperature_high_c = 70
```

7. Acceptance Checks

Step 1 Confirm that all three sector ports are up and passing traffic.

Step 2 Run a 15-minute throughput test across the backhaul and record the result in the ticket.

Step 3 Check that the router appears on the monitoring dashboard with no active alarms.

Step 4 Photograph the cabinet, the cable labels and the earth bonding.

Step 5 Hand the site over to the Network Operations Centre by phone and close the ticket.

8. Troubleshooting

Symptom	Likely cause	Action
Sector port stays down	Radio not powered, or PoE budget too low	Check the injector LED and set the PoE budget to 30 W per port
Backhaul drops every few minutes	MTU mismatch with the far end	Set the MTU to 9100 on both ends and reload the interface
No metrics on the dashboard	Collector unreachable from VLAN 999	Ping the collector from the router and raise a ticket with the NOC

If a fault cannot be cleared within two hours, roll back to the previous configuration and escalate to the on-call engineer.

9. Revision History

Revision	Date	Change
A	3 February 2026	First issue
B	19 June 2026	Microwave backhaul added
C	12 October 2026	Firmware 7.4 and telemetry section